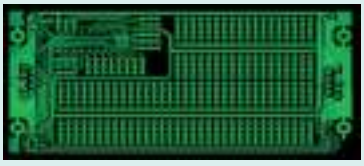
WHY MAKING SEMICON CHIPS IS HARD

From crystal growth to final testing, semiconductor manufacturing involves thousands of precision steps, advanced machinery and a deep global supply chain—making it one of the most difficult engineering challenges

1. DESIGN

Process: Engineers create the chip's circuit design using specialised Electronic Design Automation (EDA) software. Designs specify transistor layouts, logic gates and interconnects

Difficulty: Requires extreme precision, advanced architecture planning and flawless nanoscale simulation—mistakes here ripple through the entire process



2. WAFER PREPARATION

Process: Ultra-pure silicon is grown into cylindrical ingots, sliced into wafers and polished

Difficulty: Demands flawless crystal purity and atomic-level smoothness to ensure defect-free chips

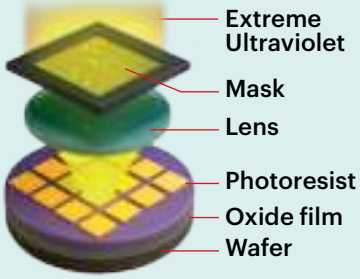


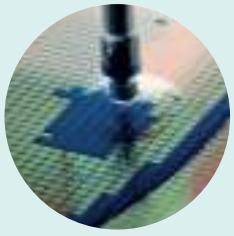
3. FABRICATION

(Front-end Processing)

Photolithography: Circuit patterns are etched onto wafers using laser light, masks and photoresist

Difficulty: Uses billion-dollar Extreme Ultraviolet (EUV) lithography machines to create features smaller than a virus



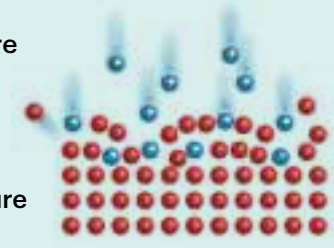


Etching: Unwanted material is chemically or physically removed to form the designed circuit patterns

Difficulty: Requires selective removal with nanometre precision without damaging adjacent structures

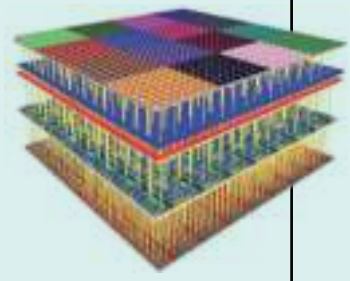
Deposition & Ion Implantation: Thin films are deposited and impurities implanted to change electrical properties

Difficulty: Atomic-level control is essential to ensure yield and performance



4. INTERCONNECT

(Back-end Processing)



Process: Dozens of microscopic metal layers and insulating layers are stacked to connect thousands of transistors

Difficulty: Aligning and insulating up to 60 layers with nanometre accuracy is a formidable engineering challenge

5. TESTING & DICING

Process: Electrical testing ensures each chip meets design specifications before the wafer is cut (diced) into individual dies

Difficulty: Requires high-speed precision testing; dicing must avoid micro-cracks that could ruin chips

